Universidade Federal Fluminense

LINKÖPINGS UNIVERSITET

Teaching Ethical Artificial Intelligence to Computer Science Students:
A Case Study Applied to a Belief-Desire-Intention Multi-Agent System

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Agenda

1. Introduction
2. Unified Ethical Principles Guide
3. Ethics Principles
4. Problems
5. Proposed Method
6. Application of the Method in a Case Study Using a BDI-Based MAS
7. Dissemination of a Survey to Gather Initial Impressions of the Proposed Method

1. Introduction

The Three Laws of Robotics, proposed by science fiction writer Isaac Asimov, were created as principles to guide the behavior of intelligent robots.

First Law

A robot may not injure a human being or, through inaction, allow a human being to come to harm.

Second Law

A robot must obey orders given it by human beings except where such orders would conflict with the First Law.

Third Law

A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

Source: [18]

1. Introduction

Ethics refers to values and principles that **prioritize human well-being** and guide the responsible design and implementation of autonomous and intelligent systems.

Source: [1]

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Ethics in AI

Subfield of applied ethics and technology focused on the ethical issues raised by the design, development, implementation, and use of AI.

Goal

Seek to maximize the benefits of AI systems for society and individuals.

Mitigation

Prevent and minimize the potential risks and harms caused by the use of AI technology.

Source: [1]

1. Introduction

Life-Cycle Integration

Ethical principles are fundamental for developing responsible AI systems and must be considered throughout their entire life cycle.

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Numerous ethical guides exist, each offering different principles and interpretations based on their origin (e.g., government, NGO, academia) and sociocultural context.

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Unified Reference Framework

Due to the lack of consensus, we have established a unified set of ethical principles as a reference to guide the development of responsible AI systems.

Source: [1]

2. Unified Ethical Principles Guide

Principles	HLEG*	UNESCO	IEEE	OECD
EP1 – Sustainable Development and Social Well-being	**Harm prevention**, **Social and Environmental Well-being**, and Fairness	Proportionality, **do no harm**, and **sustainability**	Human rights and prioritization of **well-being**	Inclusive growth, **sustainable development**, and **well-being**
EP2 – Justice and Non-Discrimination	**Fairness**, Diversity, **Non-Discrimination** and Equity	**Justice** and **non-discrimination**	–	Human rights and democratic values, including **fairness** and privacy
EP3 – Privacy and Data Protection	**Privacy** and Data Governance	Right to **privacy** and **data protection**	–	Human rights and democratic values, including fairness and **privacy**
EP4 – Responsibility and Accountability	**Responsibility**	**Responsibility** and **accountability**	**Accountability**	**Accountability**
EP5 – Robustness and Security	**Technical Robustness** and Safety	**Security** and protection	–	**Robustness**, **security**, and protection
EP6 – Human Supervision	Respect for **human autonomy**, Human Actions, and **Human Supervision**	**Human supervision** and determination	–	–
EP7 – Transparency and Explainability	**Explainability**	**Transparency** and **explainability**	**Transparency**	**Transparency** and **explainability**

Table 1: Recurring ethical principles in international guidelines

3. EP1 Principle: Sustainable Development and Social Welfare

Preservation & Well-being

AI or system must preserve **human rights** and provide individual and collective **benefits**.

It must not cause harm and should support **sustainable development goals (SDGs)**.

Monitoring & Assessment

The economic and cultural **impacts** caused by AI should be continuously monitored.

Continuous assessment is vital to prevent undesirable consequences throughout its lifecycle.

Sources: [1] [2]

3. Examples of MAS Focused on Promoting Sustainable Development and Social Welfare

Smart Agriculture

Optimization of water resources and precise fertilizer application through autonomous agent monitoring.

Urban Mobility

Traffic signal control and public transport management to enhance traffic flow and reduce congestion.

Sustainable Finance

Stock trading negotiations favoring companies with sustainable practices and divesting from non-certified entities.

Sources: [3, 4, 5, 6] MAS Research Papers

3. EP2 Principle: Fairness and Non-Discrimination

Social Justice & Equity

AI and system must promote **social justice**, **equity**, and **non-discrimination** in accordance with international law.

It should be accessible to all, considering specific needs and **diversities** of different **social groups** (e.g., age, culture, language, disability).

Prevention of Bias

It is fundamental that AI does not reinforce or **perpetuate discriminatory** or **biased outcomes** related to race, gender, religion, or any other characteristic.

The goal is to ensure a fair distribution of benefits and keep individuals free from prejudice.

Source: [2]

3. Example of MAS Focused on Promoting the Principle of Justice and Non-Discrimination

Judicial Process Management

MAS were applied in the management of judicial processes, assisting in workload distribution and hearing anticipation, which contributes to improved productivity and faster access to justice.

Source: [9]

3. EP3 Principle: Privacy and Data Protection

Rights & Protection

Covers the **rights of data subjects** regarding the processing of their **personal data** within the AI context.

- Data consent management
- Sensitive data protection techniques
- Data leakage prevention

Legal Framework

Compliance with regulations like the **General Data Protection Regulation (GDPR)** is essential throughout the AI lifecycle.

Ensures subjects can exercise their rights and provide **informed consent** for data usage.

Source: [10]

3. Examples of MAS Focused on Promoting Privacy and Data Protection

Data Consent

Agents must adapt and operate in compliance with data protection standards in a jurisdiction, such as Brazil or Europe.

- Transition from LGPD to GDPR compliance
- Context-aware regulatory adaptation

Negotiation Process

The goal is to minimize sensitive information sharing between agents without compromising agreement possibilities.

- Secure multi-party computation logic
- Differential privacy in agent interaction

Sources: [13, 14, 15]

3. EP4 Principle: Responsibility and Accountability

Responsibility

This principle deals with the attribution of responsibility for decisions made by AI to the corresponding actors according to their role in the life cycle. In MAS, responsibility can be directed towards the **assignment** of **degrees** of responsibility among agents.

Accountability

Accountability refers to the ability of an agent or system to **justify** or **inform** about its actions for which it has responsibility to another agent legitimized to make this demand.

Sources: [2, 11, 12]

3. Examples of MAS and Adoption of the Responsibility and Accountability Principle

Degrees of Responsibility

Each agent holds a portion of responsibility for the success or failure of the task.

- Challenges in assigning different degrees of responsibility to agents.

Accountability Process

Accountability within the negotiation process ensures agents justify their actions.

- Agents must provide justification if they violate commitments made during negotiations.

Sources: [16, 17, 18]

3. EP5 Principle: Robustness and Safety

Robustness & Reliability

Ensures AI or system functions as **intended** even under **adverse** and **unpredictable** conditions.

Resilience & Safety

Mechanisms for **prevention** or **mitigation** of failures must minimize unexpected harm. AI must remain **resilient** against **cyberattacks** or **vulnerabilities** throughout its lifecycle.

Sources: [1]

3. Examples of MAS and Adoption of the Robustness and Safety Principle

Secure Communication

Secure communication between agents, utilizing a certificate authority service.

- Ensures encryption and identity verification.

Industrial Resilience

System performance in the face of industrial failures, combined with accountability strategies applied to industrial contexts.

- Focuses on fault tolerance and justification of actions.

Sources: [19, 20, 21, 22]

3. EP6 Principle: Human Oversight

This principle states that AI users, especially in applications that can affect people's lives and safety, must not be **subject to decisions made exclusively by AI**, which restricts human autonomy.

AI must place the human being at the **center** of decision-making, enabling **oversight** and **control** of the operations performed.

Sources: [1]

3. Examples of MAS and Adoption of the Human Oversight Principle

Military Domain Scenario

In a military domain scenario, agents can undergo or request **human intervention**.

- For instance, when an agent detects enemy movement, it may request **human assistance** to handle the situation.

Sources: [23]

3. EP7 Principle: Transparency and Explainability

Transparency

Focuses on **information disclosure** (objectives and limitations) so that stakeholders understand the AI system.

Allows those affected to **challenge results** by understanding the logic, data, and processes behind decisions.

Explainability

Provides **intelligible information** regarding inputs, outputs, and internal functioning.

AI algorithms and outcomes must be fundamentally **explainable** and **auditable**.

Sources: [1]

4. Problems



Identification

How can negative ethical consequences in systems be identified?

4. Problems

Identification

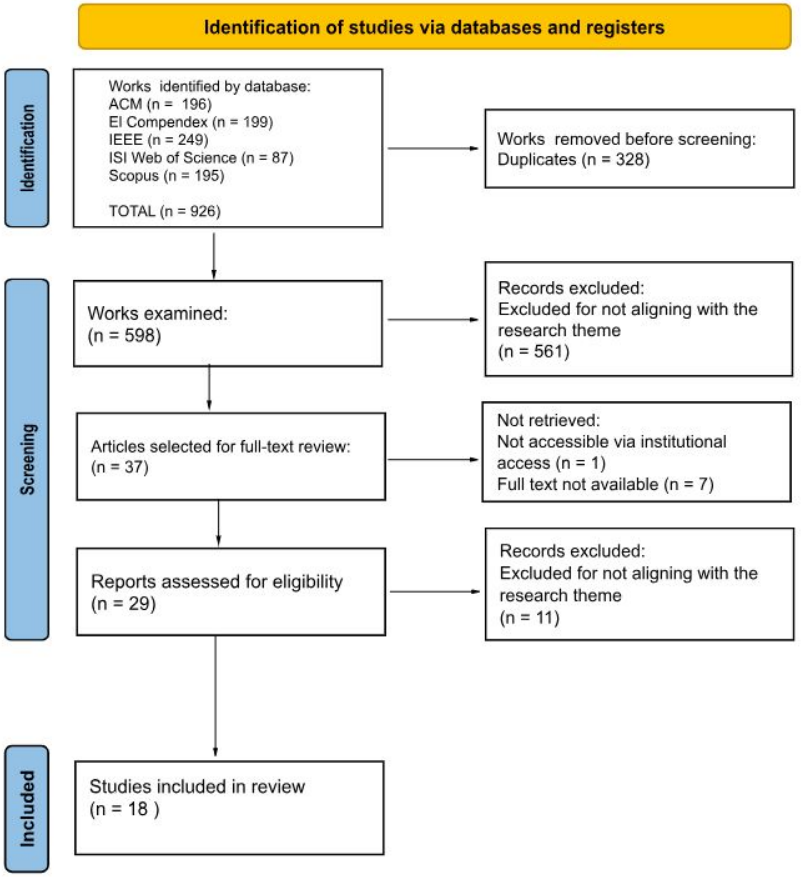
How can negative ethical consequences in systems be identified?

Resolution

How can such ethical consequences be addressed?

4. Problems

In a literature review, we observed that there are both theoretical and practical activities aimed at developing ethical thinking.



4. Problems

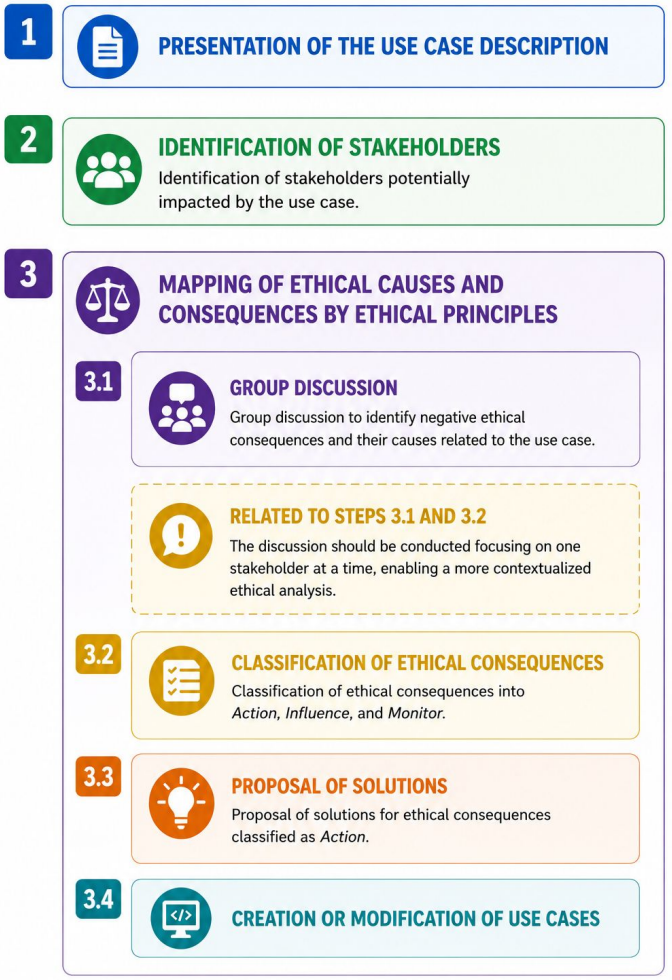
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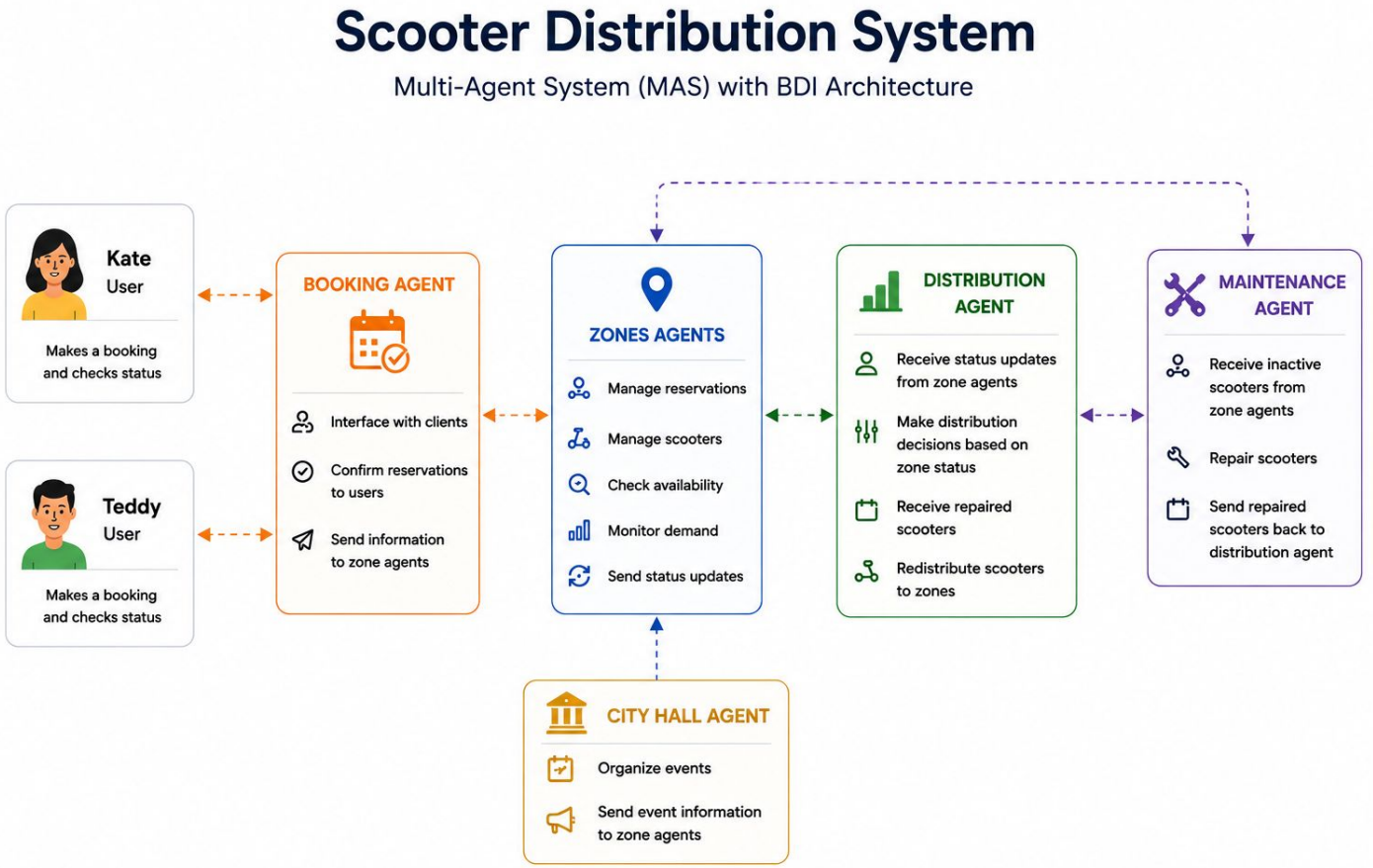
Activities	Identification of Impacted Stakeholders	Identification of Negative Ethical Consequences	Identification of Causes Associated with Ethical Consequences	Classification of consequences into actionable, influenceable, or monitorable.	Proposal of Solutions for Actionable Consequences	Creation or Modification of Use Cases
Consequence Scanning		X		X	X	
Diagram Construction		X				
Discussions	X	X				
Ethical Analysis through Classical Ethical Theories		X			X	
Hands-on Development Activities					X	X
Literature-Based Activities		X			X	
Metacognition	X	X				
Proposed Method	X	X	X	X	X	X

5. Proposed Method



6. Application of the Method in a Case Study

A BDI-based Multi-Agent System (MAS) for Scooter Management was used as a case study to demonstrate the application of the proposed method. The study was conducted by a group composed of one researcher and two graduate students.



Step 1: Use Case Description

Use Case 1 - Redistribute Scooters Between City Zones
Objective: Ensure that different areas of the city maintain an adequate number of available electric scooters through a Multi-Agent System (MAS) composed of autonomous agents responsible for monitoring and redistributing scooters.
Agents involved: • Distributor • Zones

Step 1: Use Case Description

Use Case 1 - Redistribute Scooters Between City Zones

Main Flow

1. Each zone sends its current status containing the following information:
   - zone type;
   - available scooters;
   - demand level;
   - unmet demand;
   - inactive scooters;
   - local events;
   - user satisfaction in the zone.
2. The Distribution Agent receives updated status information from all zones.
3. The Distribution Agent analyzes the status of all zones and decides whether scooters should be transferred to one or more priority zones.
4. The Distribution Agent calculates the number of scooters to be transferred to a zone based on the annual usage history of each zone.
5. Each Zone Agent receives a calculated quantity of scooters from the Distribution Agent's central stock.

AF1 - Lack of Available Scooters in the Distribution Agent's Central Stock

Alternative Flow for Step 10 of the Main Flow

If there are not enough scooters available in the Distribution Agent's Central Stock:

1. The Distribution Agent checks whether there are zones with available scooters and low current demand.
2. If so, it transfers available scooters from the identified zone to a priority zone

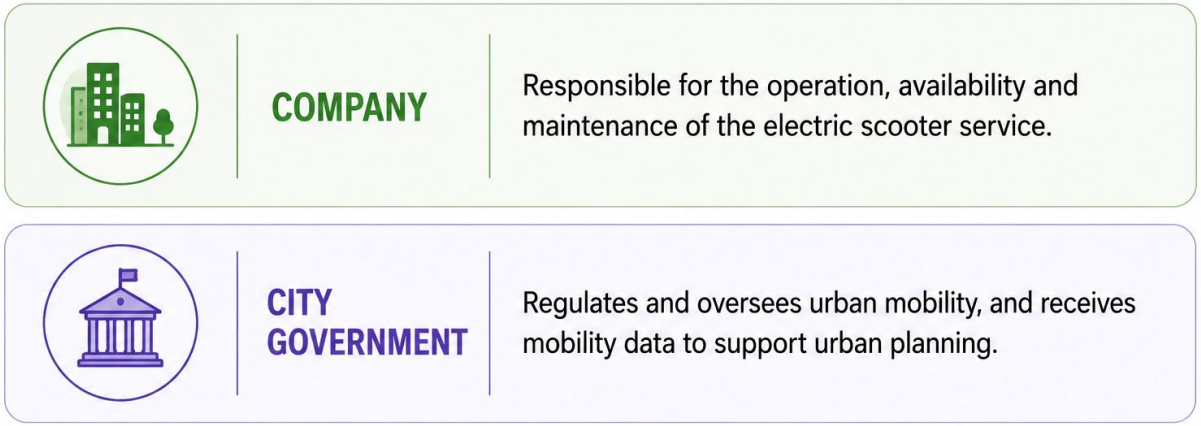
Step 2: Identification of Impacted Stakeholders



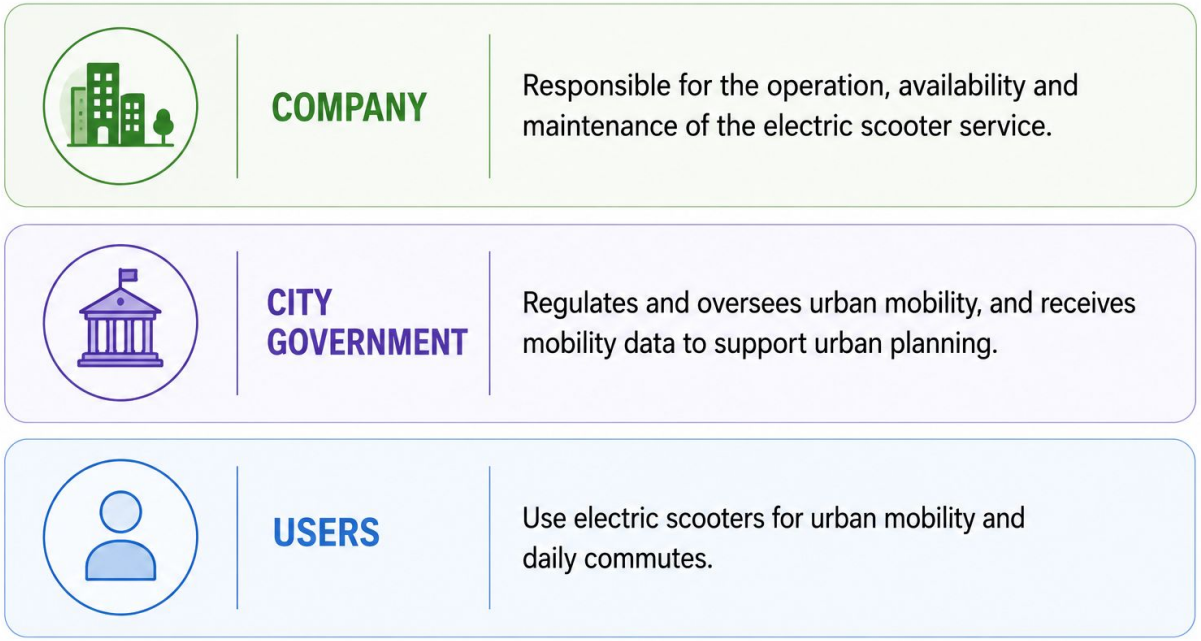
COMPANY

Responsible for the operation, availability and maintenance of the electric scooter service.

Step 2: Identification of Impacted Stakeholders



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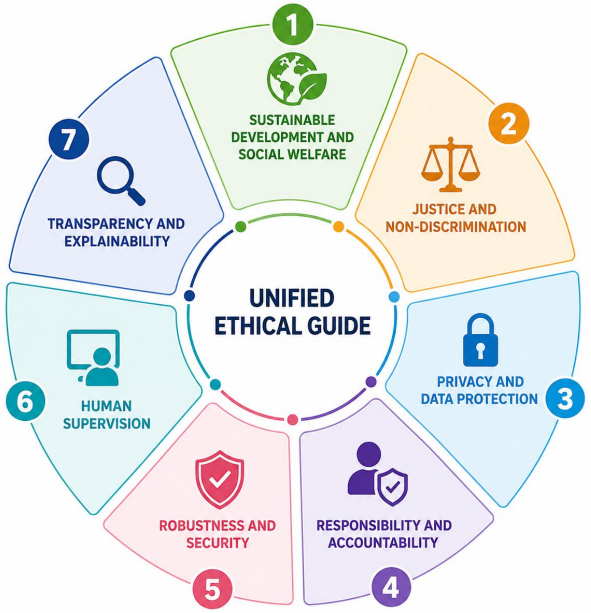


Step 3: Mapping of Ethical Causes and Consequences by Ethical Principle

Step 1: Group Activity

Group discussions were conducted to identify negative consequences and their causes associated with the use case and the ethical principles defined in the unified guideline.

The Unified Ethical Guide was used as the basis for identifying negative ethical consequences.



Trigger Ethics Questions were also used to stimulate reflection on potential negative ethical consequences in the system.



Step 3: Mapping of Ethical Causes and Consequences

Use Case 1 - Redistribute Scooters Between City Zones

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Justice and Non-Discrimination

Zones with historically low scooter availability may continue receiving fewer scooters, even if the demand in those areas changes over time.

Step 3: Mapping of Ethical Causes and Consequences

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Sustainable Development and Social Welfare

An area may be left without available scooters if another zone is prioritized.

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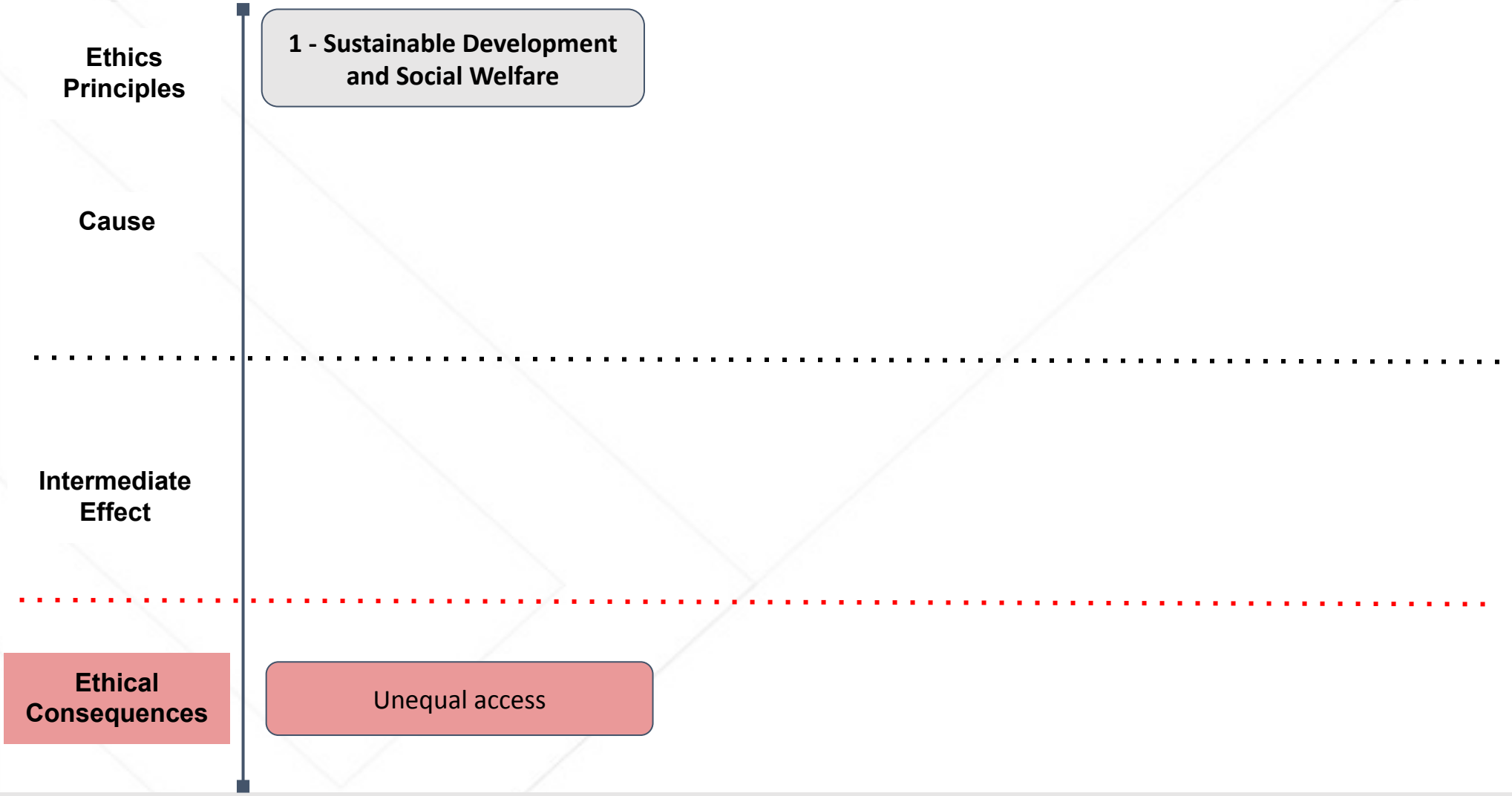
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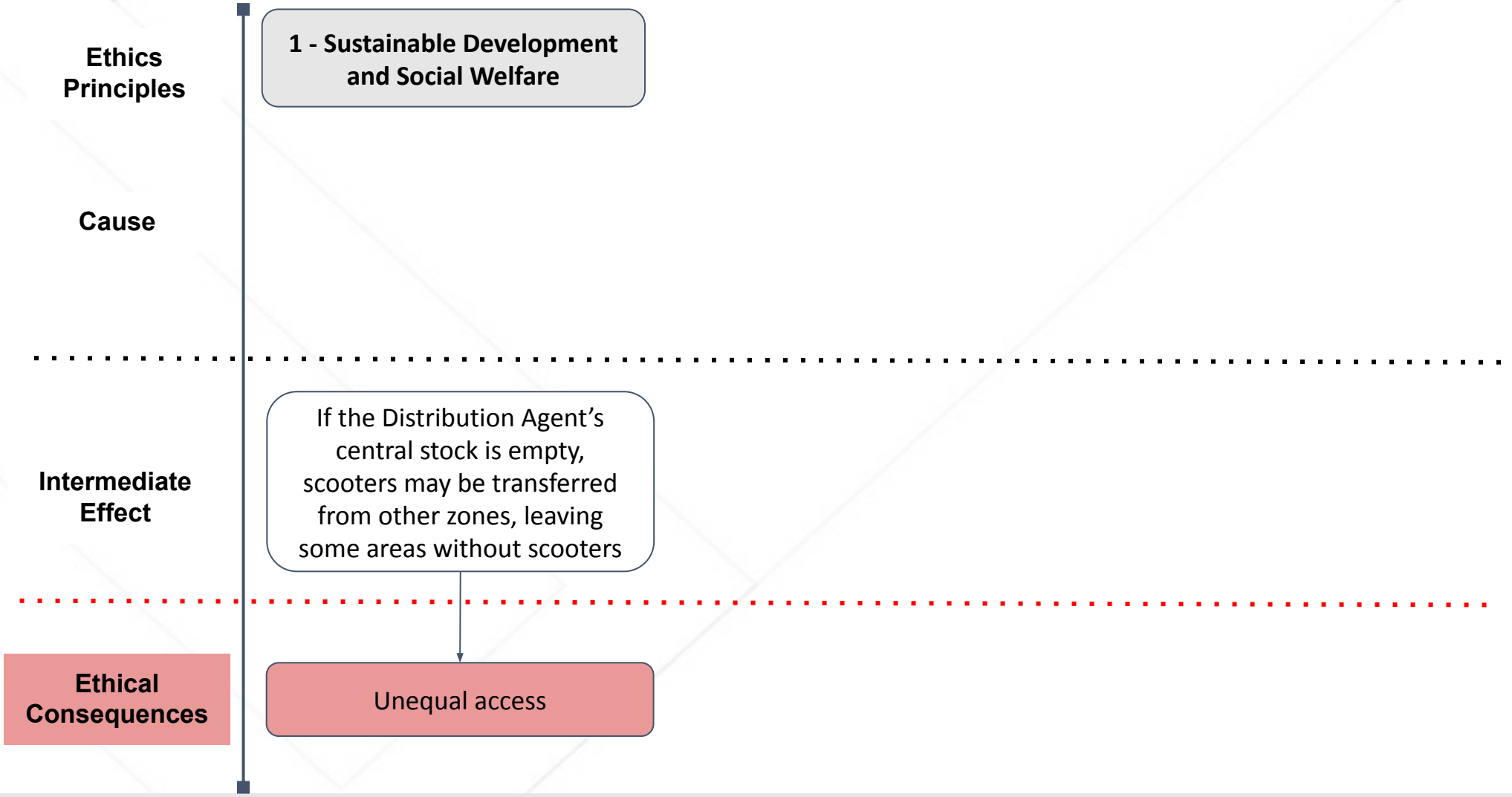
Privacy and Data Protection

To calculate the demand of each zone at a given time, it is necessary to collect user data such as location and usage times

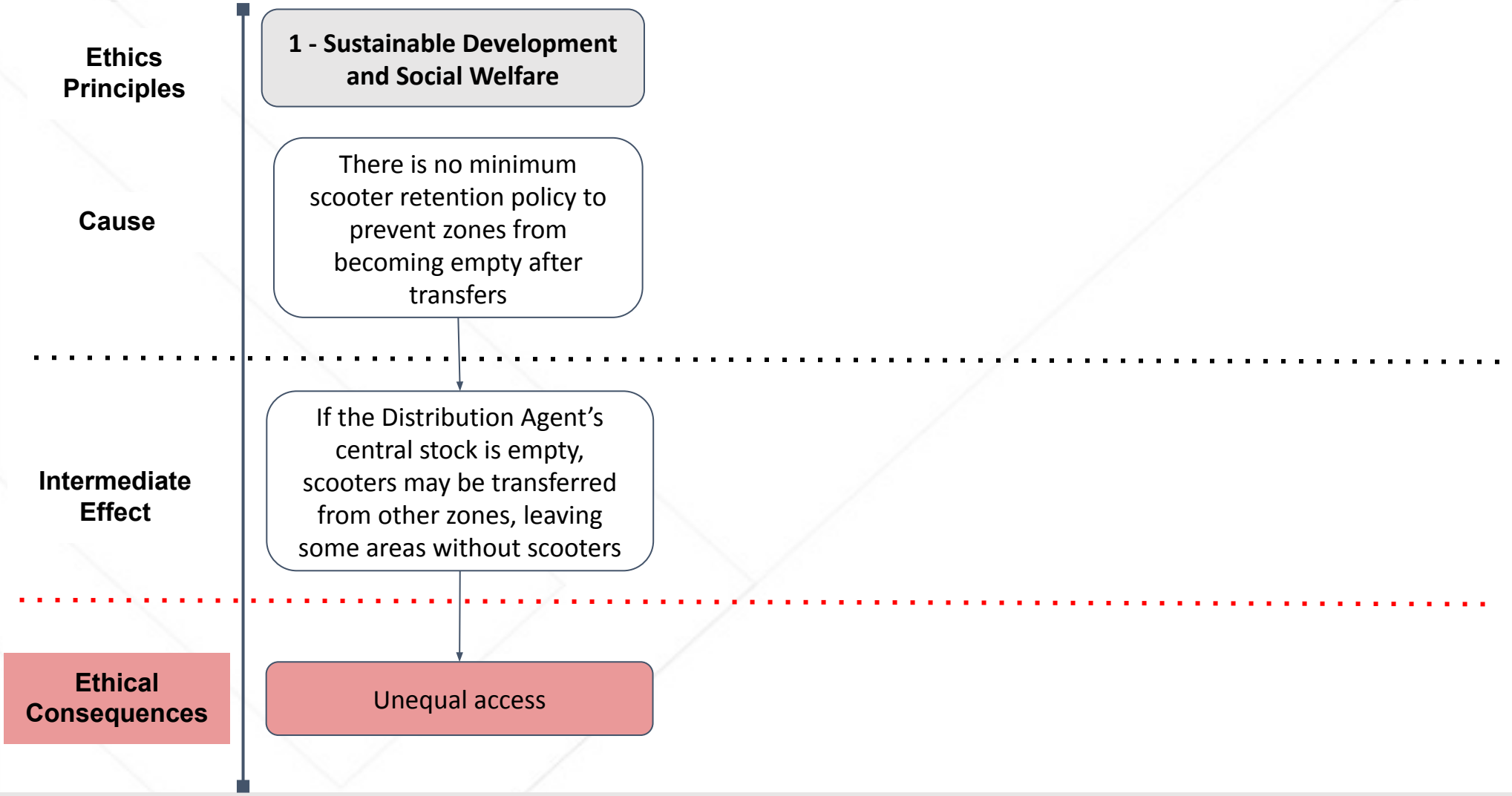
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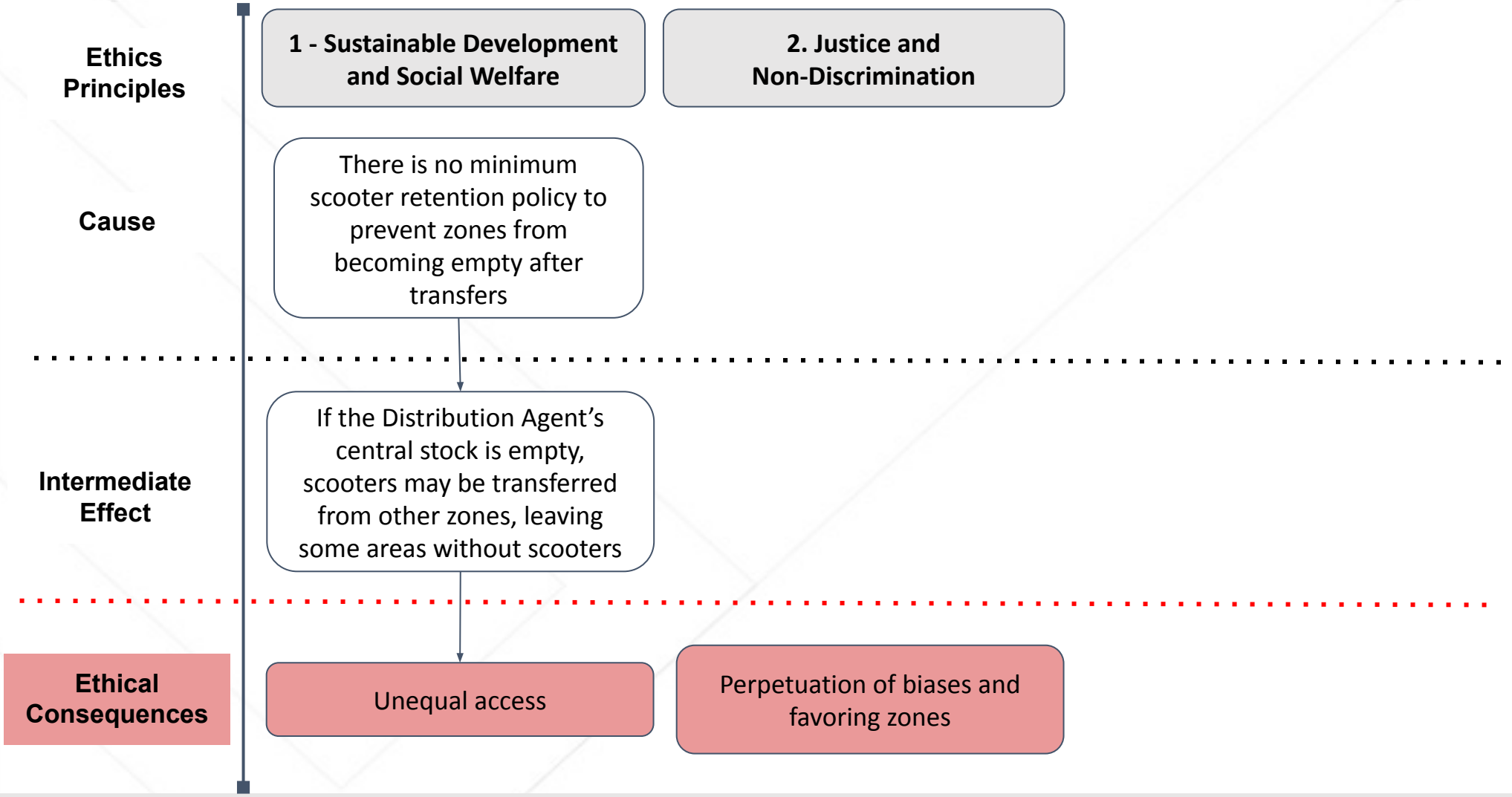
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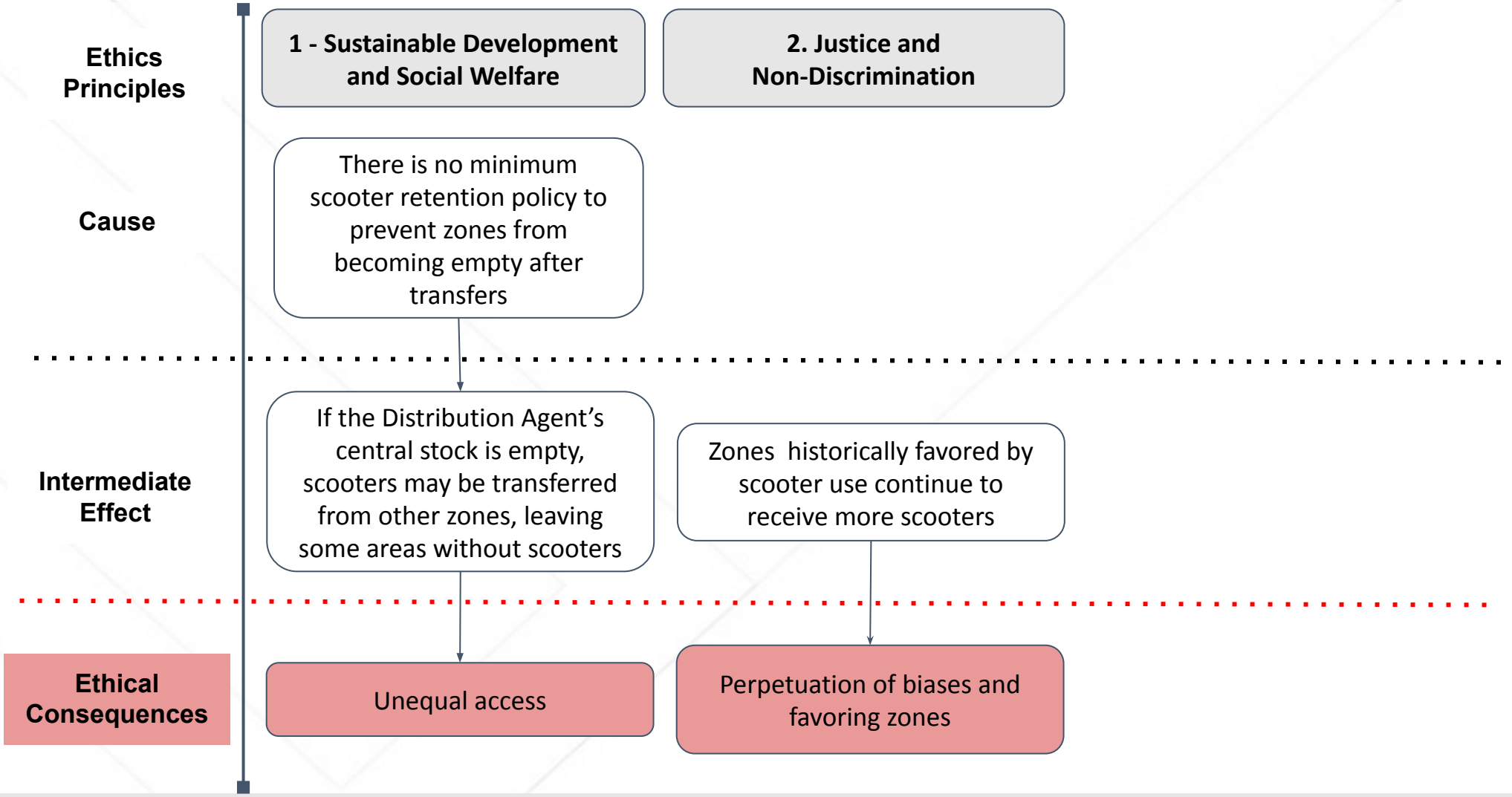
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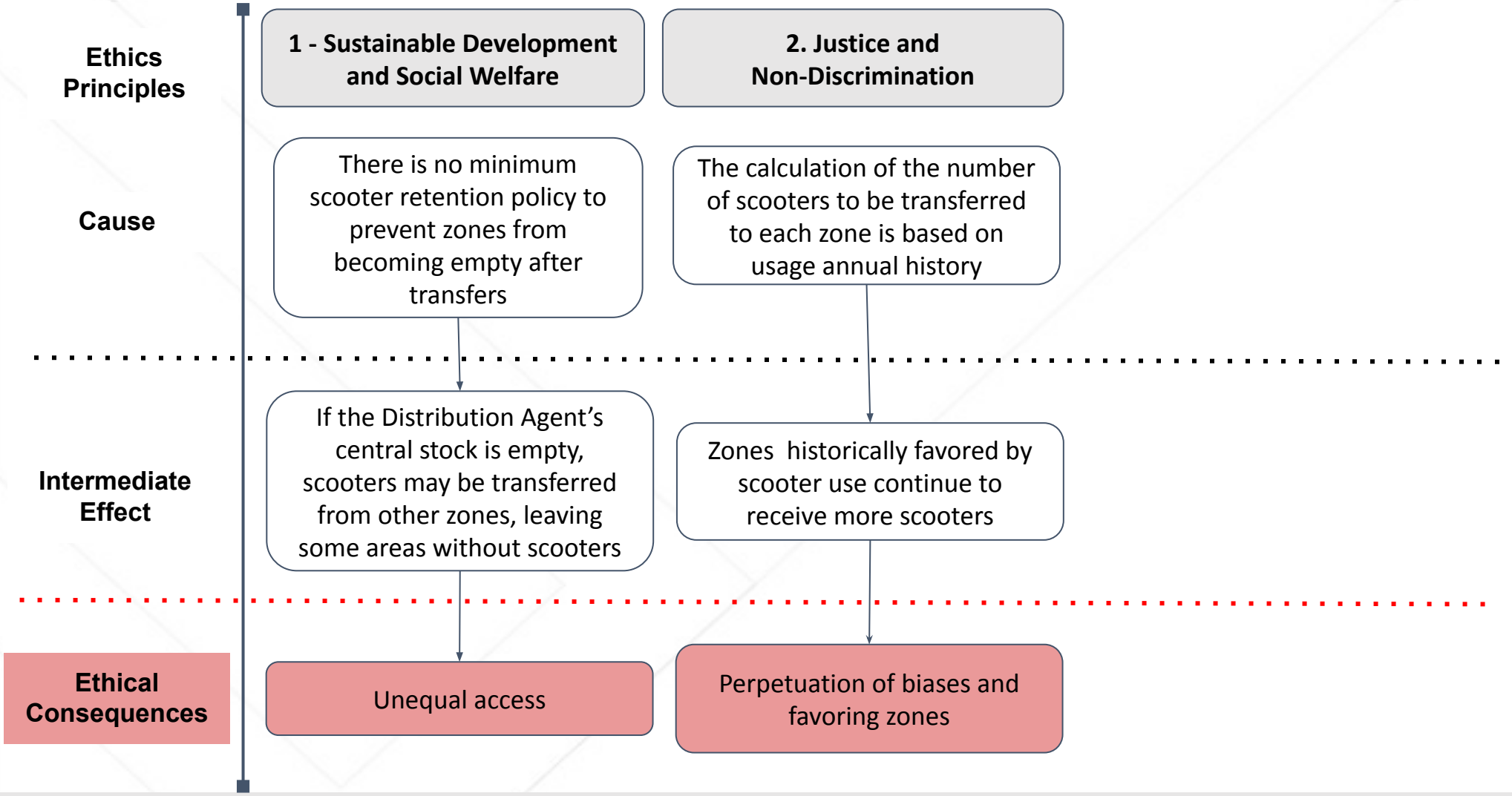
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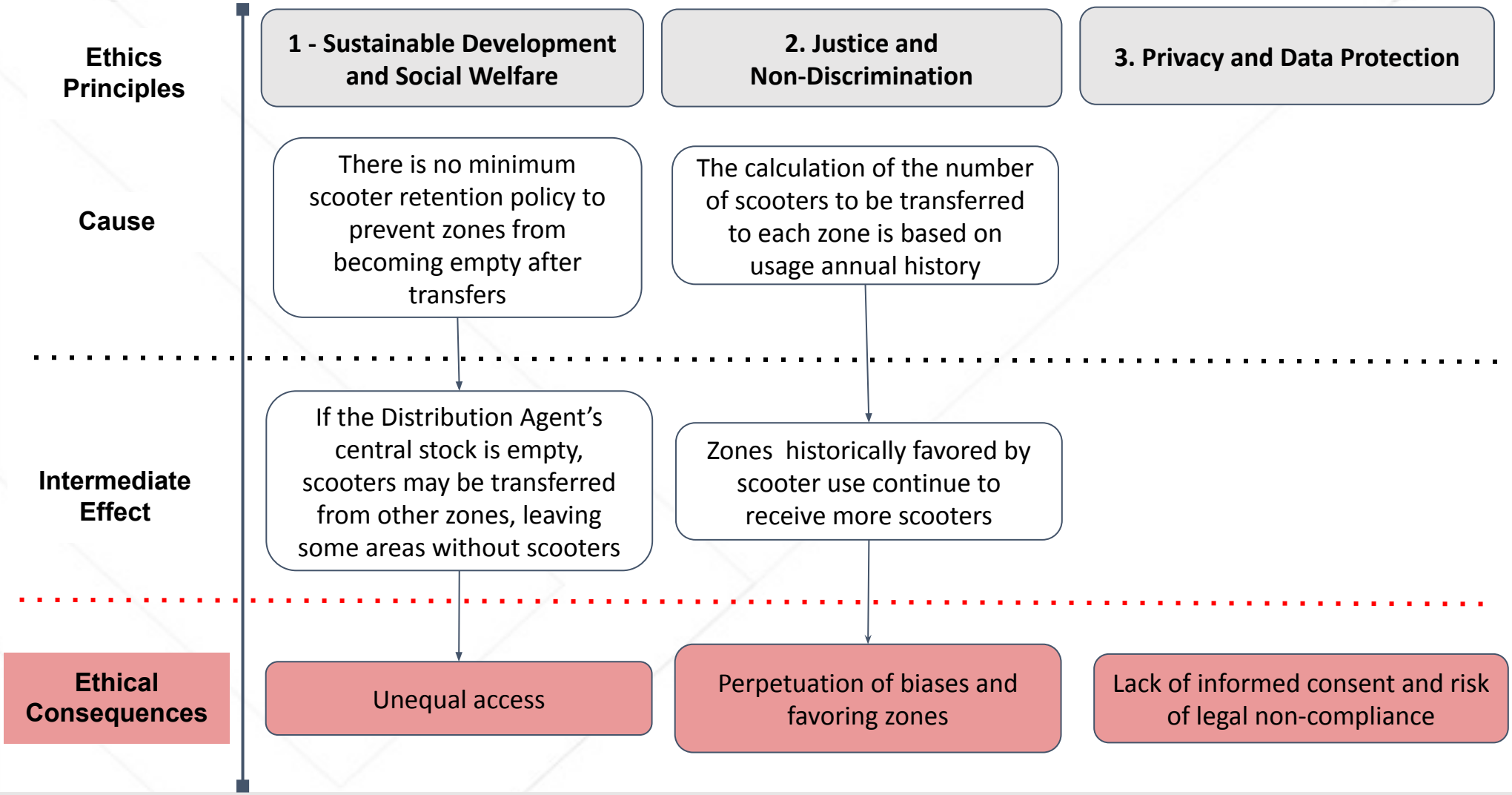
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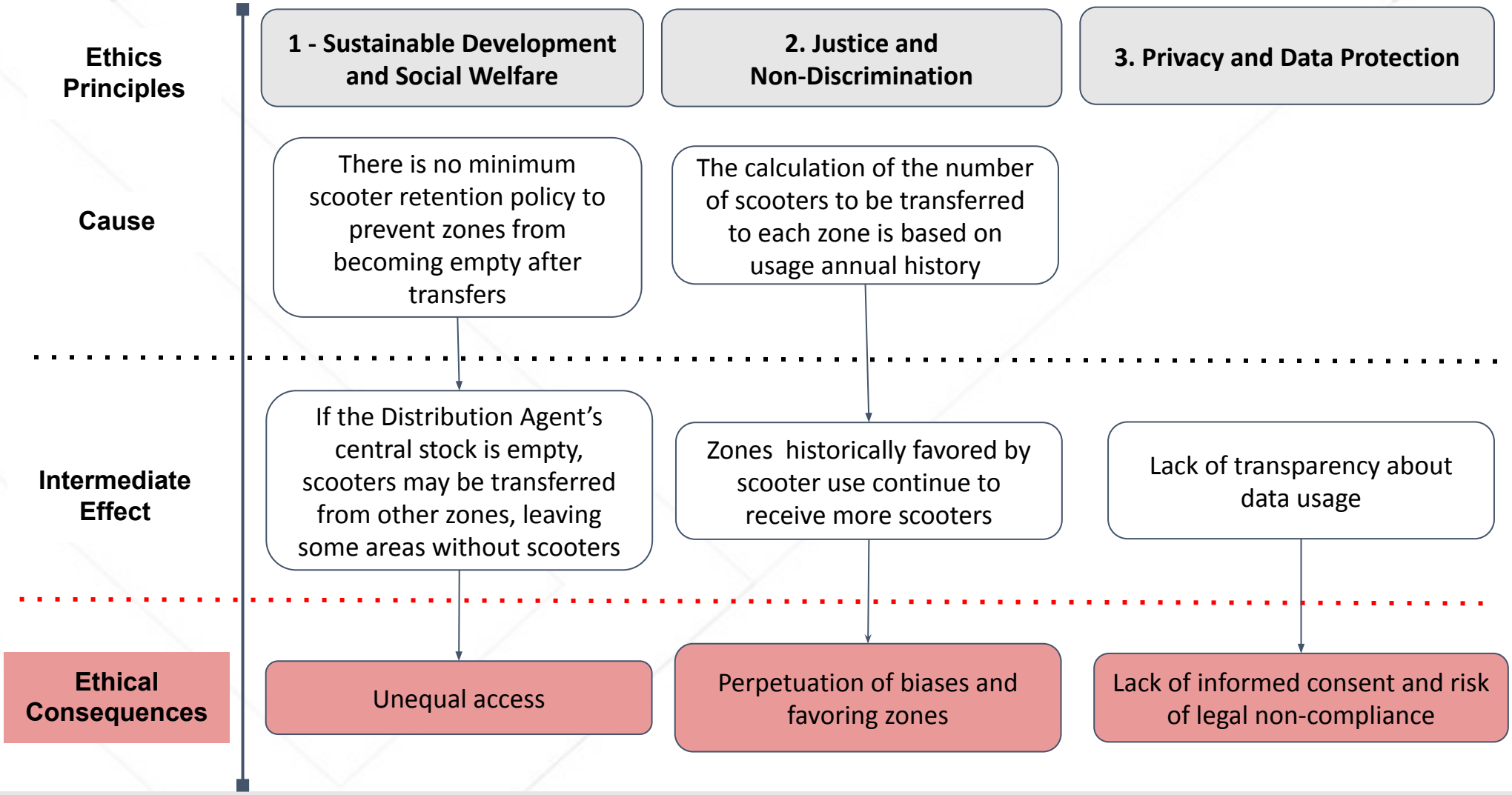
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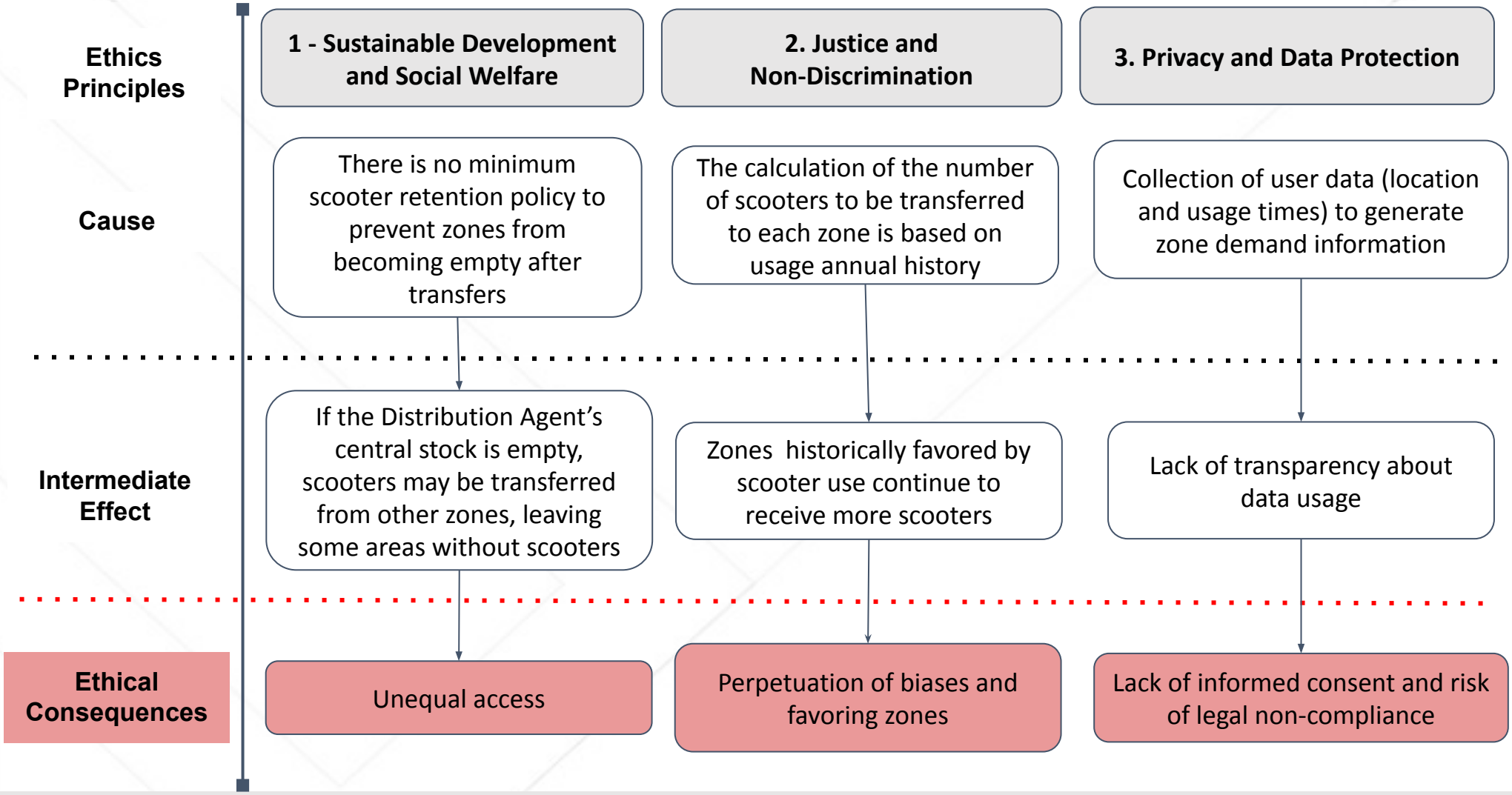
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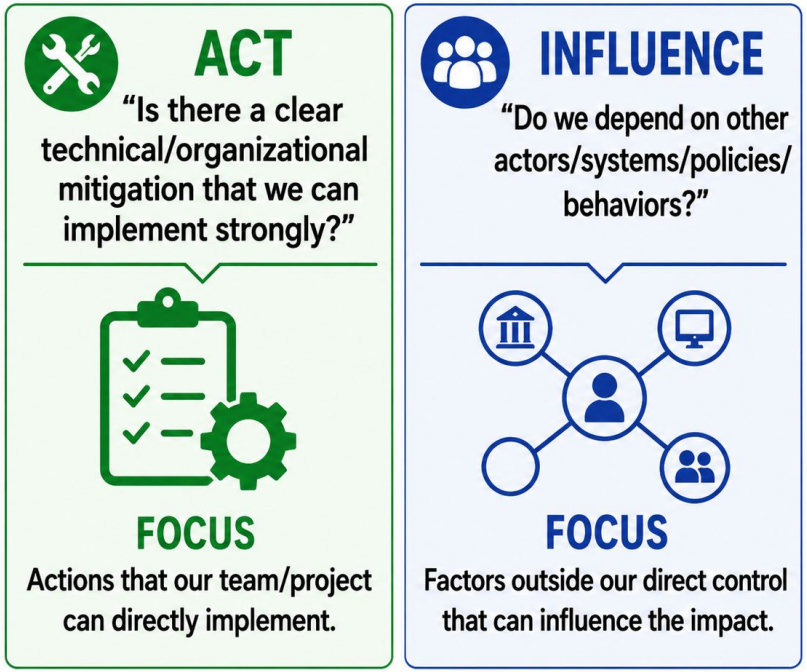
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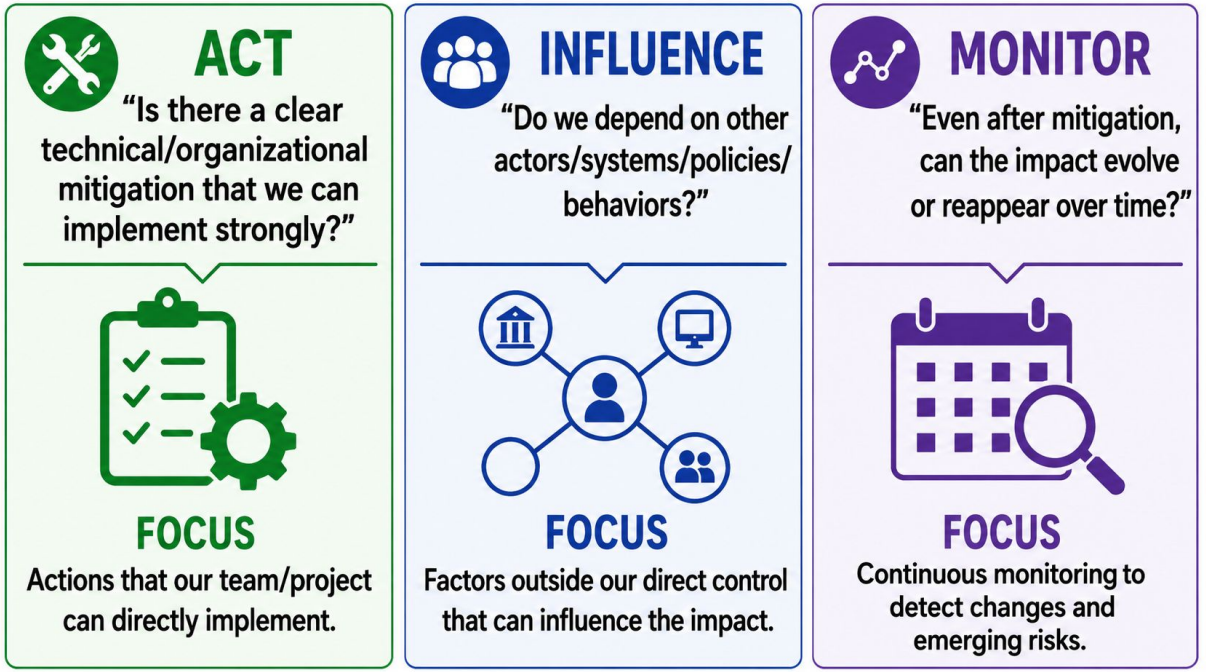
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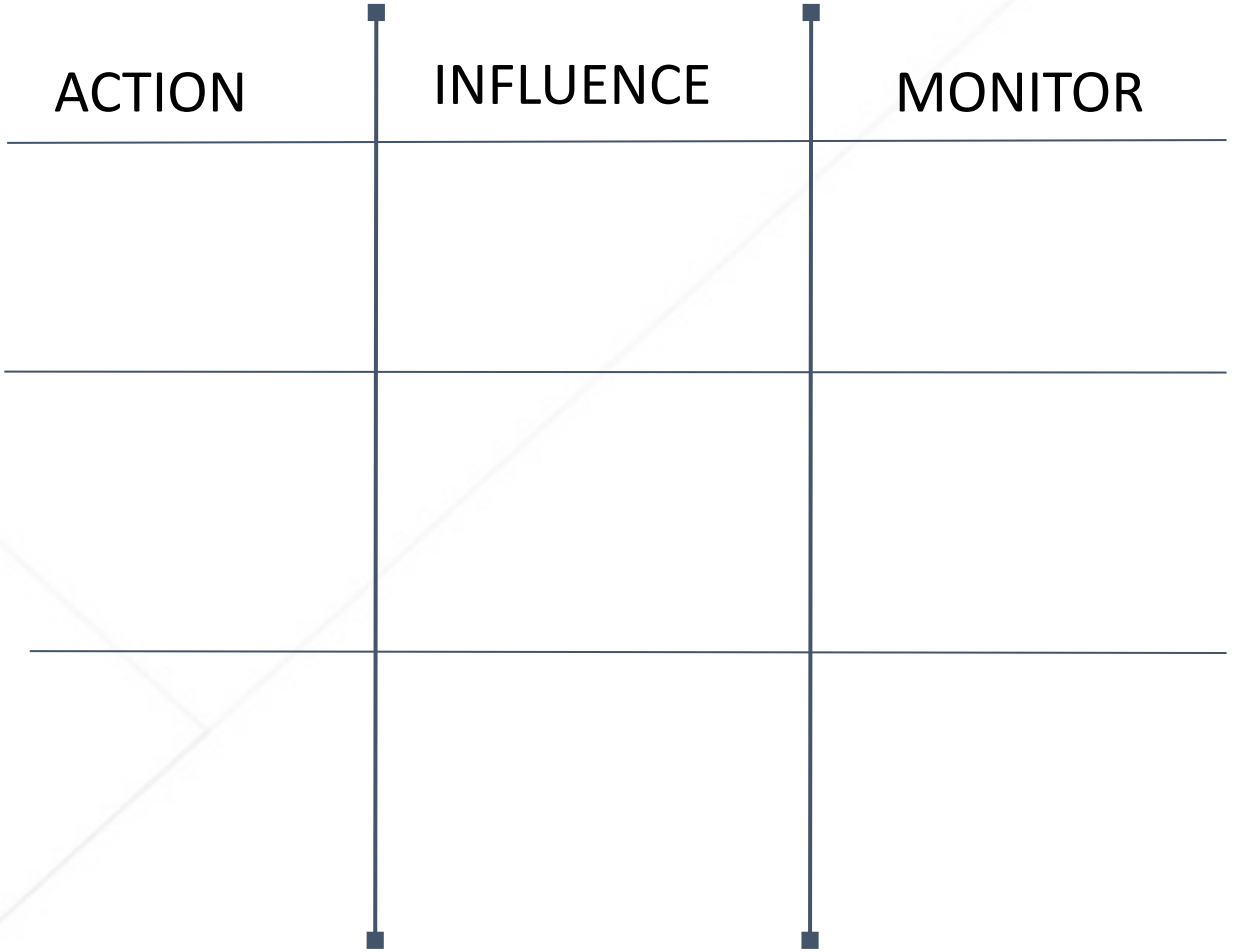


Step 3.2 : Classification of ethical negative consequences into Action, Influence, and Monitor



Step 3.3: Classification of ethical consequences into Action, Influence, and Monitor

Cause	Consequences
There is no minimum scooter retention policy to prevent zones from becoming empty after transfers	Unequal access
The calculation of the number of scooters to be transferred to each zone is based on usage annually history	Perpetuation of biases and favoring zones
Collection of user data (location and usage times) to generate zone demand information	Lack of informed consent and risk of legal non-compliance



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ACTION	INFLUENCE	MONITOR
Define a minimum scooter threshold per zone to avoid complete depletion after transfers		

Step 3.3: Classification of ethical consequences into Action, Influence, and Monitor

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ACTION	INFLUENCE	MONITOR
Define a minimum scooter threshold per zone to avoid complete depletion after transfers		Continuously monitor whether all zones are receiving adequate numbers of scooters.

Step 3.3: Classification of ethical consequences into Action, Influence, and Monitor

Cause	Consequences	ACTION	INFLUENCE	MONITOR
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The calculation of the number of scooters to be transferred to each zone is based on usage annually history	Perpetuation of biases and favoring zones	Consider repressed demand and monthly demand when calculating scooter distribution among zones		
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Collection of user data (location and usage times) to generate zone demand information	Lack of informed consent and risk of legal non-compliance	Adjust the app's consent request to explicitly inform users about the types of data collected and their intended purposes.		

Step 3.3: Classification of ethical consequences into Action, Influence, and Monitor

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Collection of user data (location and usage times) to generate zone demand information	Lack of informed consent and risk of legal non-compliance	Adjust the app's consent request to explicitly inform users about the types of data collected and their intended purposes.	Adjust organizational guidelines to ensure clear consent mechanisms	

Step 3.4: Creation or Modification of Use Cases

ACTION
1. Define a minimum scooter threshold per zone to avoid complete depletion after transfers
2. Consider repressed demand and monthly demand when calculating scooter distribution among zones
3. Adjust the app's consent request to explicitly inform users about the types of data collected and their intended purposes.

Step 3.4: Creation or Modification of Use Cases

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3. Adjust the app's consent request to explicitly inform users about the types of data collected and their intended purposes.

Adjusting the description of the current requirement
If so, it transfers available scooters from the identified zone to a priority zone while maintaining a minimum scooter threshold in the source zone.

Step 3.4: Creation or Modification of Use Cases

ACTION
1. Define a minimum scooter threshold per zone to avoid complete depletion after transfers
2. Consider repressed demand and monthly demand when calculating scooter distribution among zones
3. Adjust the app's consent request to explicitly inform users about the types of data collected and their intended purposes.

Adjusting the description of the current requirement
If so, it transfers available scooters from the identified zone to a priority zone while maintaining a minimum scooter threshold in the source zone.

Adjusting the description of the current requirement
The Distribution Agent calculates the number of scooters to be transferred to each zone based on the zone's **monthly** usage history and r**epressed demand.**

Step 3.4: Creation or Modification of Use Cases

ACTION
1. Define a minimum scooter threshold per zone to avoid complete depletion after transfers
2. Consider repressed demand and monthly demand when calculating scooter distribution among zones
3. Adjust the app's consent request to explicitly inform users about the types of data collected and their intended purposes.

Adjusting the description of the current requirement

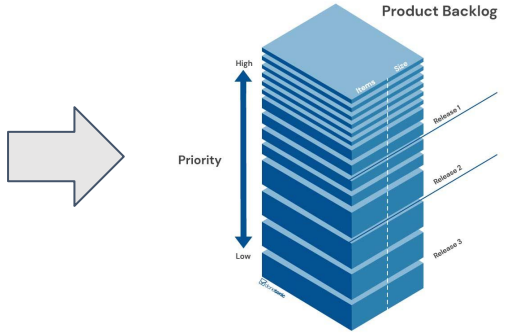
If so, it transfers available scooters from the identified zone to a priority zone while maintaining a minimum scooter threshold in the source zone.

Adjusting the description of the current requirement

The Distribution Agent calculates the number of scooters to be transferred to each zone based on the zone's **monthly** usage history and r**epressed demand.**

New requirement

A new requirement will be created and added to the development backlog.



Step 3.4: Creation or Modification of Use Cases

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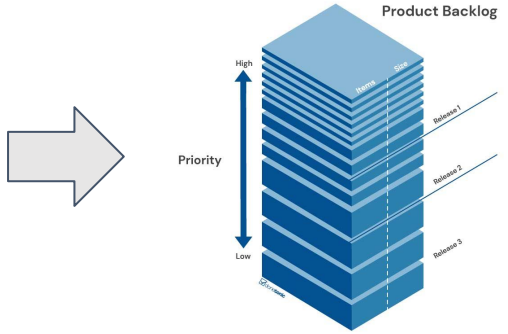
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New requirement

A new requirement will be created and added to the development backlog.



7. Dissemination of a Survey to Gather Initial Impressions of the Proposed Method



Your input is valuable!

We are conducting a survey to gather initial impressions about the proposed method for identifying and addressing negative ethical consequences in AI-based or software systems.

The questionnaire is brief, and your responses will be used exclusively for academic and research purposes.

We appreciate your participation!

Participation link: https://forms.gle/vFJEkox4tcVCETt39

Bibliography

1. COMMISSION, European Union. Ethics Guidelines for Trustworthy AI. [S. l.: s. n.], jun. 2018. Accessed on April 23, 2025. Disponível em: <https://digitalstrategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai>.

2. UNESCO. Recommendation on the Ethics of Artificial Intelligence. [S. l.: s. n.], nov. 2021. Accessed on April 23, 2025. Disponível em: <https://unesdoc.unesco.org/ark:/48223/pf0000381137.locale=en>

3. IKIDID, Abdelouafi; EL, Fazziki Abdelaziz; SADGAL, Mohamed. Internet of things and agent-based system to improve water use efficiency in collective irrigation. Computer Science and Information Systems, v. 20, n. 1, p. 405–421, 2023

4. GUO, Jiaying; CHENG, Long; WANG, Shen. CoTV: Cooperative control for traffic light signals and connected autonomous vehicles using deep reinforcement learning. IEEE Transactions on Intelligent Transportation Systems, IEEE, v. 24, n. 10, p. 10501–10512, 2023.

5. GUO, Yida; ZHANG, Cheng. Real-time Railway Transit Management Based on Multi-Agent System (MAS). In: IOP PUBLISHING, 1. JOURNAL of Physics: Conference Series. [S. l.: s. n.], 2021. v. 1828, p. 012045.

6. COINTE, Nicolas; BONNET, Grégory; BOISSIER, Olivier. Ethics-based cooperation in multi-agent systems. In: ADVANCES in Social Simulation: Looking in the Mirror. [S. l.: s. n.], 2020. P. 101–116

7. PRIYADARSHANA, HVV et al. Developing multi-agent based micro-grid management system in jade. In: IEEE. 2019 2nd International Conference on Power and Embedded Drive Control (ICPEDC). [S. l.: s. n.], 2019. P. 552–556

8. CARDOSO, Rafael C; FERRANDO, Angelo; DENNIS, Louise A et al. Implementing ethical governors in BDI. In: SPRINGER. INTERNATIONAL Workshop on Engineering Multi-Agent Systems. [S. l.: s. n.], 2021. P. 22–41.

Bibliography

9. DI MARTINO, Beniamino; ESPOSITO, Antonio; COLUCCI CANTE, Luigi. Multi agents simulation of justice trials to support control management and reduction of civil trials duration. Journal of Ambient Intelligence and Humanized Computing, Springer, v. 14, n. 4, p. 3645–3657, 2023.

10. EUROPEAN PARLIAMENT AND OF THE COUNCIL. General Data Protection Regulation (GDPR)). [S. l.: s. n.], abr. 2016. Official Journal of the European Union. Accessed on April 23, 2025. Disponível em: <https://eur-lex.europa.eu/eli/reg/2016/679/oj>.

11. YAZDANPANAH, Vahid; DASTANI, Mehdi et al. Multiagent task coordination as task allocation plus task responsibility. In: SPRINGER. MULTI-AGENT Systems and Agreement Technologies: 17th European Conference, EUMAS 2020, and 7th International Conference, AT 2020, Thessaloniki, Greece, September 14-15, 2020, Revised Selected Papers 17. [S. l.: s. n.], 2020. P. 571–588.

12. DIGNUM, Virginia. Responsible artificial intelligence: how to develop and use AI in a responsible way. [S. l.]: Springer, 2019. v. 2156.

13. CARDOSO, Paulo Henrique; CORREIA, Fernando et al. A Normative Multiagent Approach to Represent Data Regulation Concerns. In: ICAART (1). [S. l.: s. n.], 2023. P. 330–337.

14. ARHADI, Farzaneh; JENNINGS, Nicholas R. A faithful mechanism for incremental multi-agent agreement problems with self-interested and privacy-preserving agents. SN Computer Science, Springer, v. 2, n. 4, p. 272, 2021

15. XIE, Shangyu et al. Privacy preserving distributed energy trading. In: IEEE. 2020 IEEE 40th International Conference on Distributed Computing Systems (ICDCS). [S. l.: s. n.], 2020. P. 322–332.

16. YAZDANPANAH, Vahid; STEIN, Sebastian et al. Applying strategic reasoning for accountability ascription in multiagent teams, 2021.

17. YAZDANPANAH, Vahid; DASTANI, Mehdi et al. Multiagent task coordination as task allocation plus task responsibility. In: SPRINGER. MULTI-AGENT Systems and Agreement Technologies: 17th European Conference, EUMAS 2020, and 7th International Conference, AT 2020, Thessaloniki, Greece, September 14-15, 2020, Revised Selected Papers 17. [S. l.: s. n.], 2020. P. 571–588.

Bibliography

18. Asimov, Isaac. "Three laws of robotics." *Asimov, I. Runaround* 2.3 (1941).

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2026